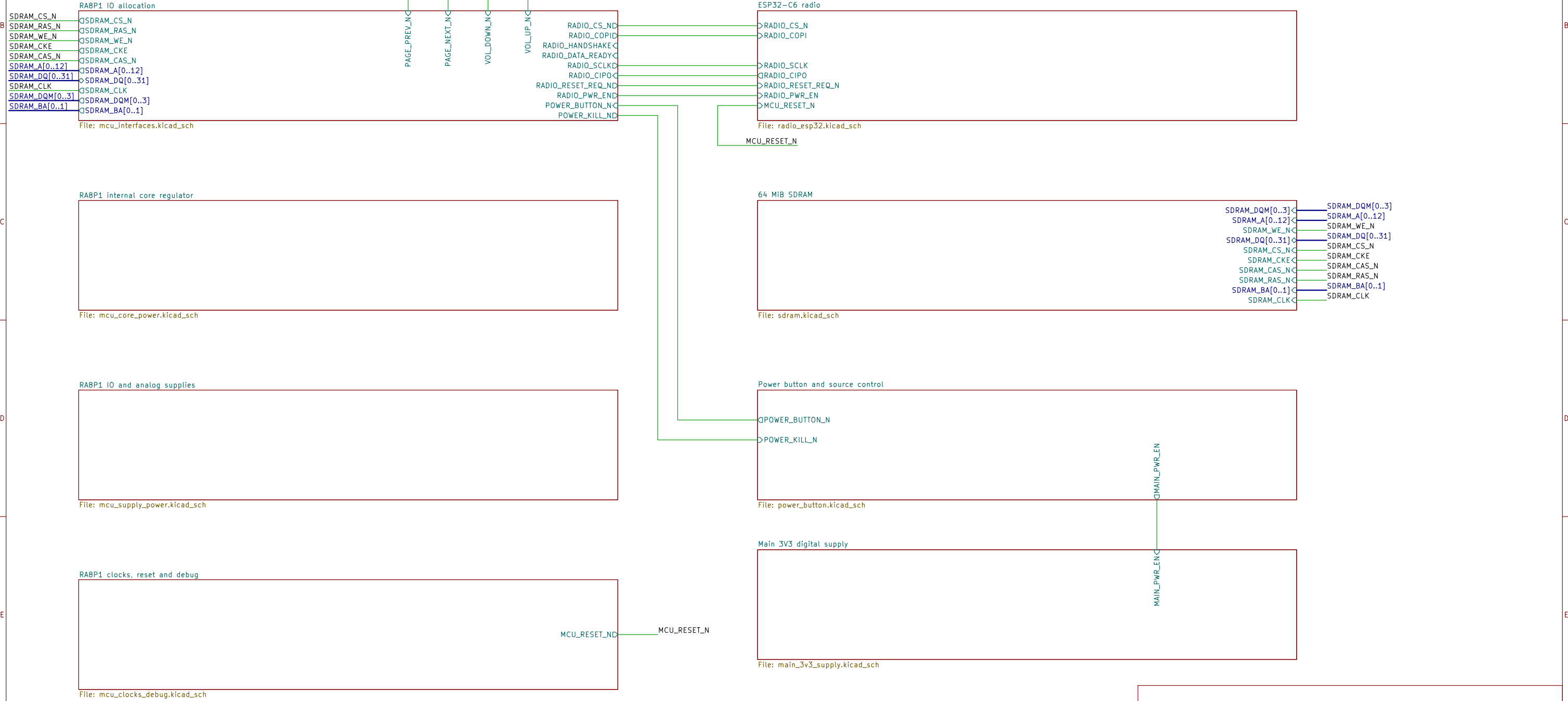


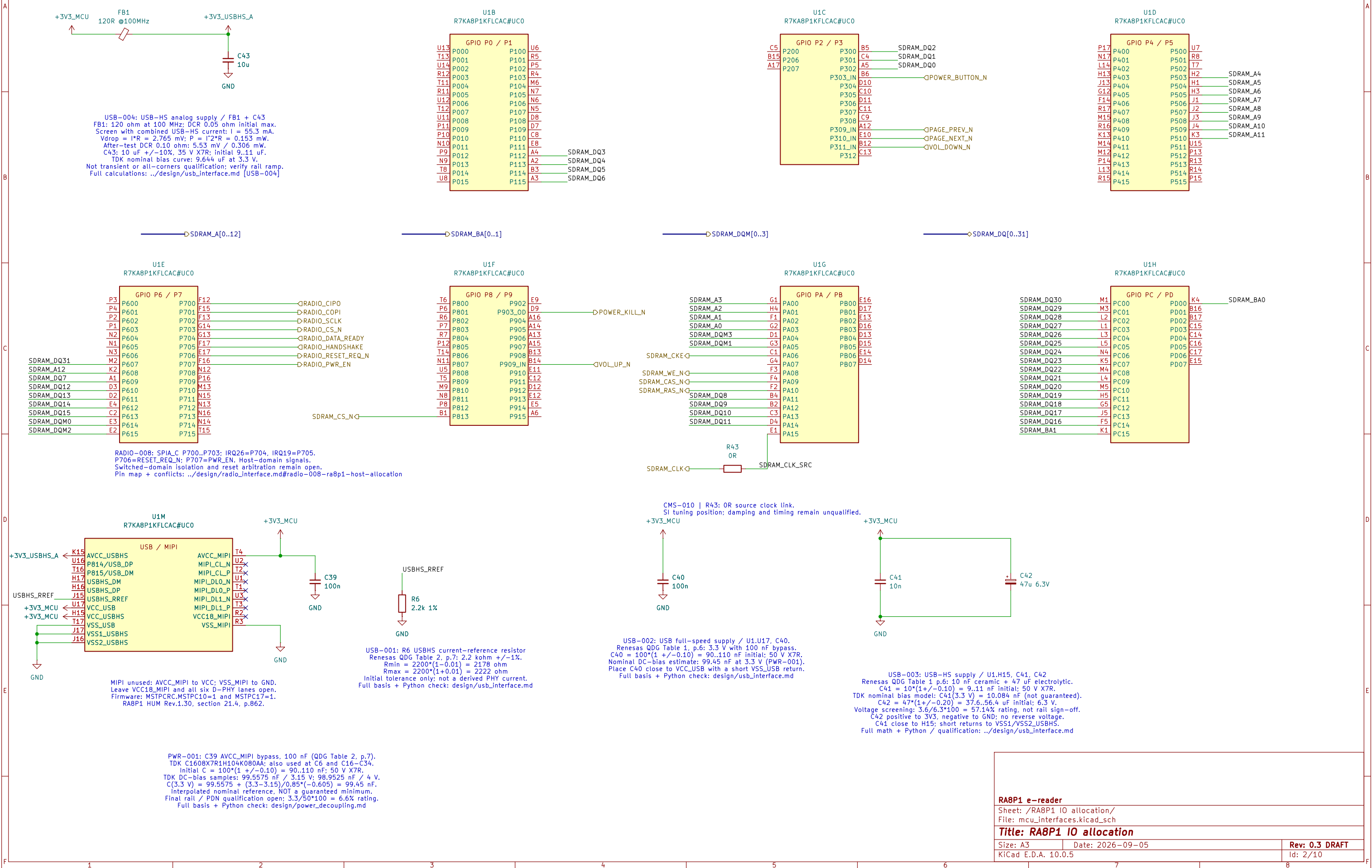
RA8P1 E-READER

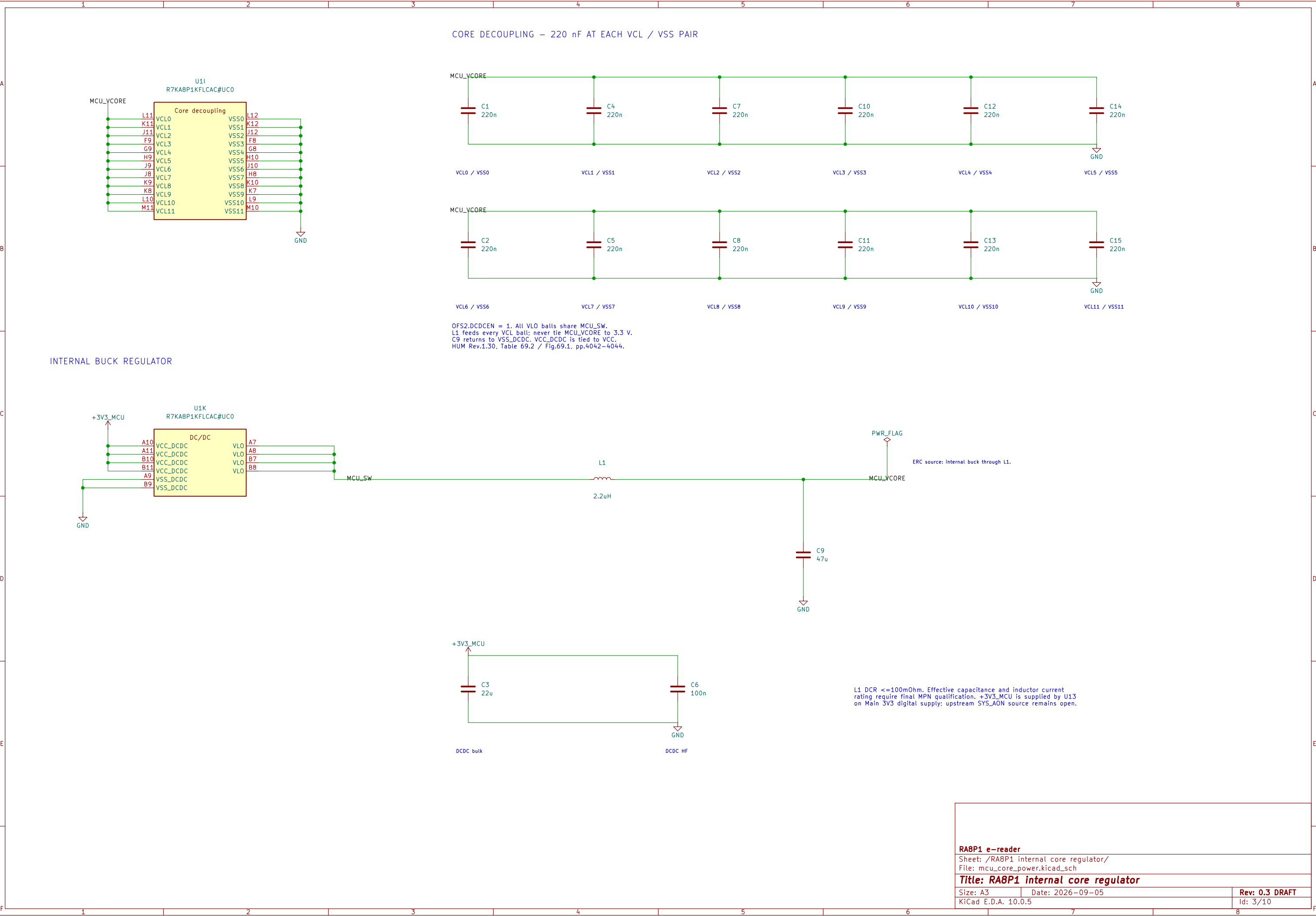
Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

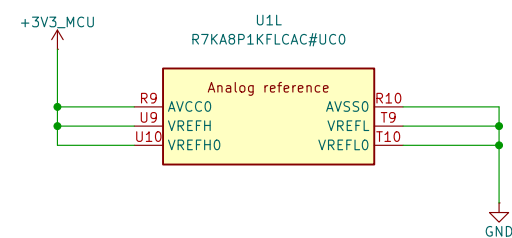
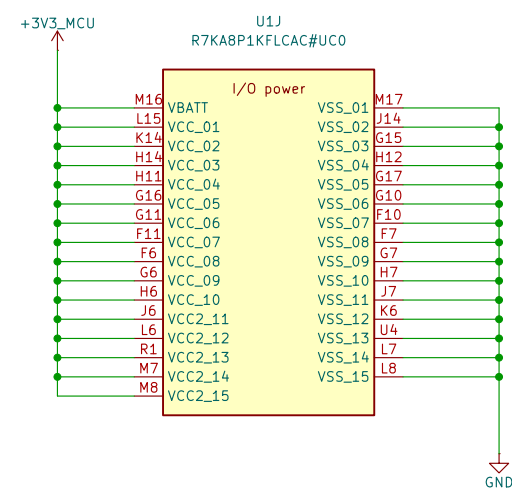


Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

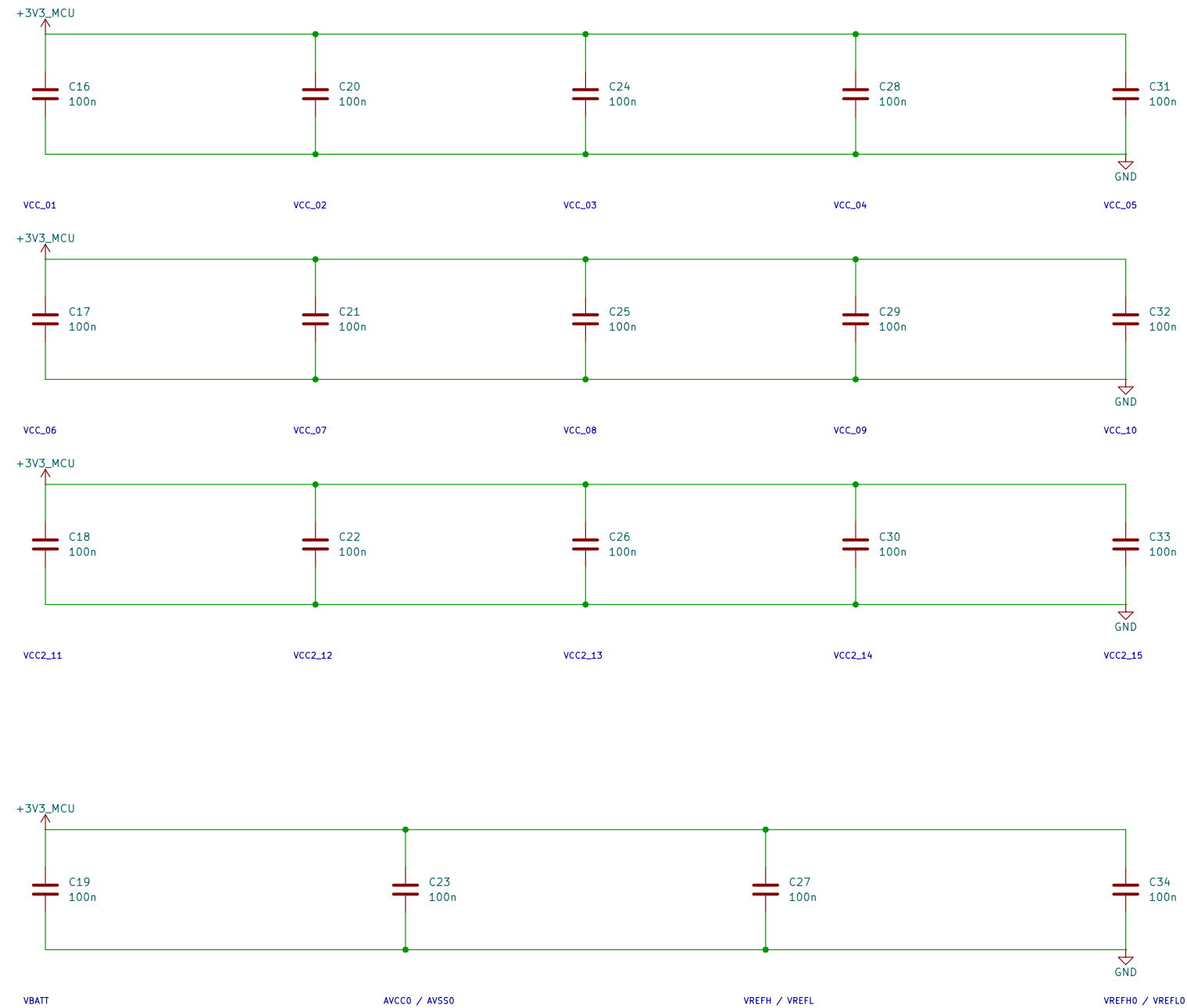
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



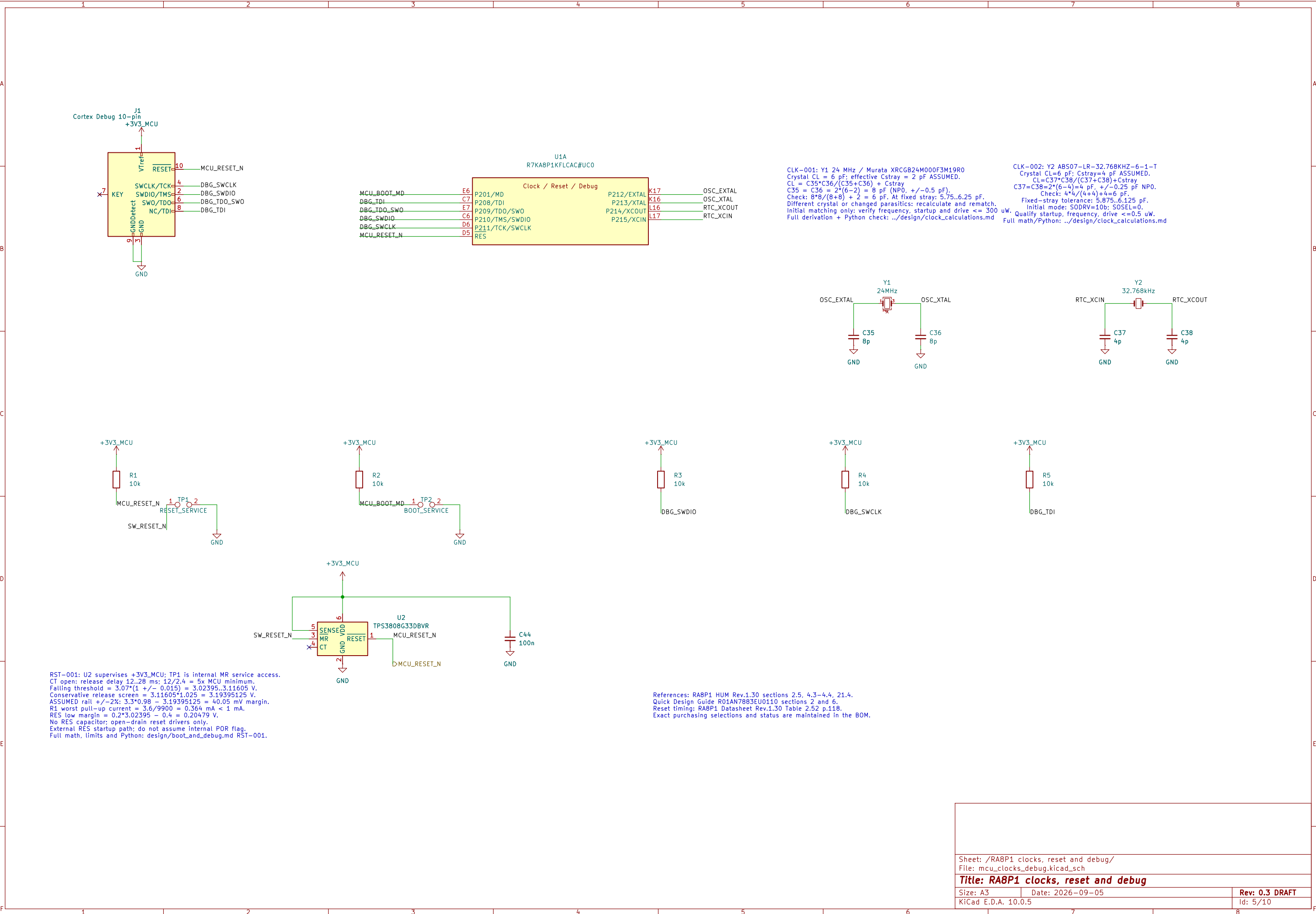




I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862. This is an electrical draft; capacitor MPN qualification remains open.

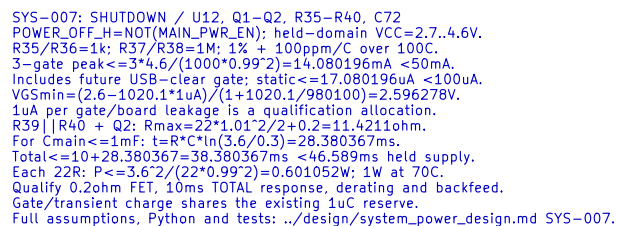
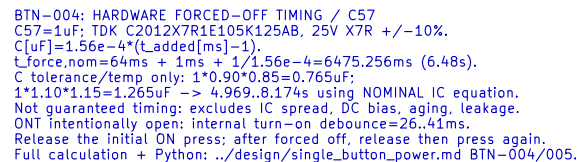


CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

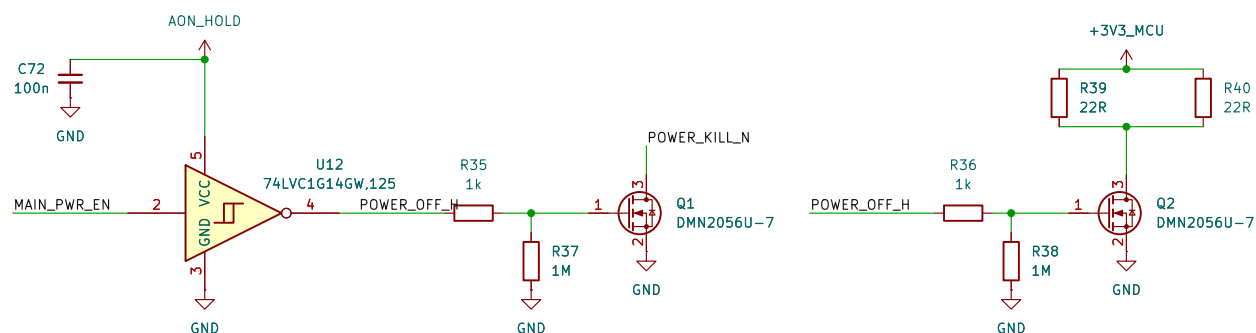
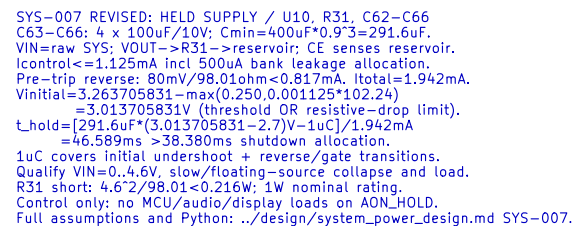
CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

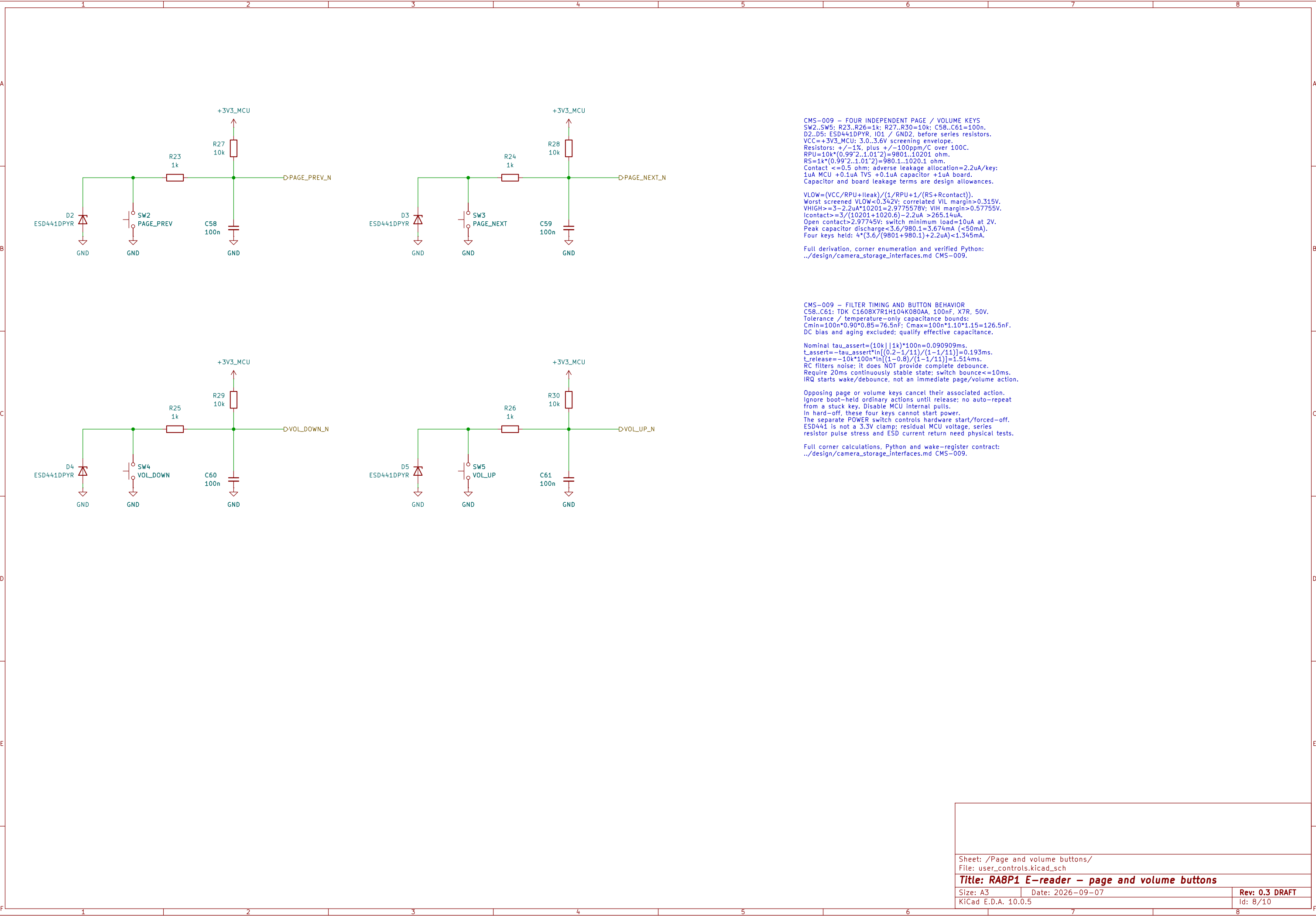
RST-001: U2 supervises +3V3_MCU; TP1 is internal MR service access.
CT open: release delay 12..28 ms; 12/2.4 = 5x MCU minimum.
Falling threshold = 3.07*(1 +/- 0.015) = 3.02395..3.11605 V.
Conservative release screen = 3.11605*1.025 = 3.19395125 V.
ASSUMED rail +/-2%: 3.3*0.98 - 3.19395125 = 40.05 mV margin.
R1 worst pull-up current = 3.6/9900 = 0.364 mA < 1 mA.
RES low margin = 0.2*3.02395 - 0.4 = 0.20479 V.
No RES capacitor; open-drain reset drivers only.
External RES startup path; do not assume internal POR flag.
Full math, limits and Python: design/boot_and_debug.md RST-001.

References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

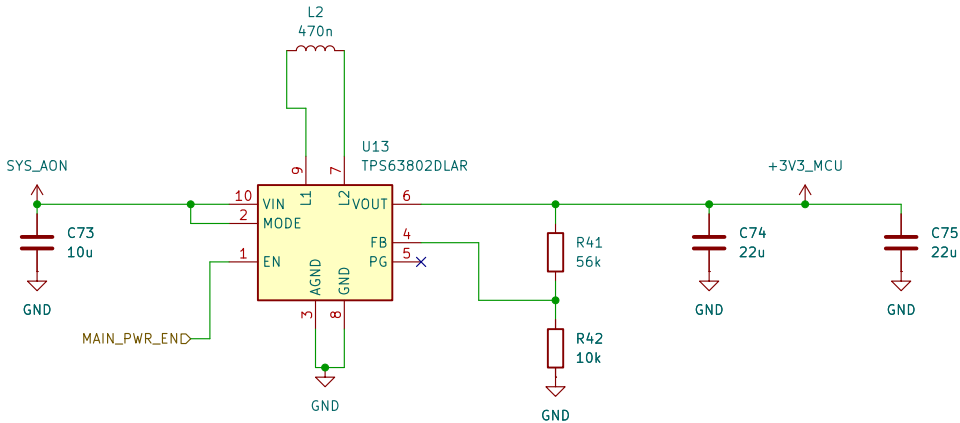


ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





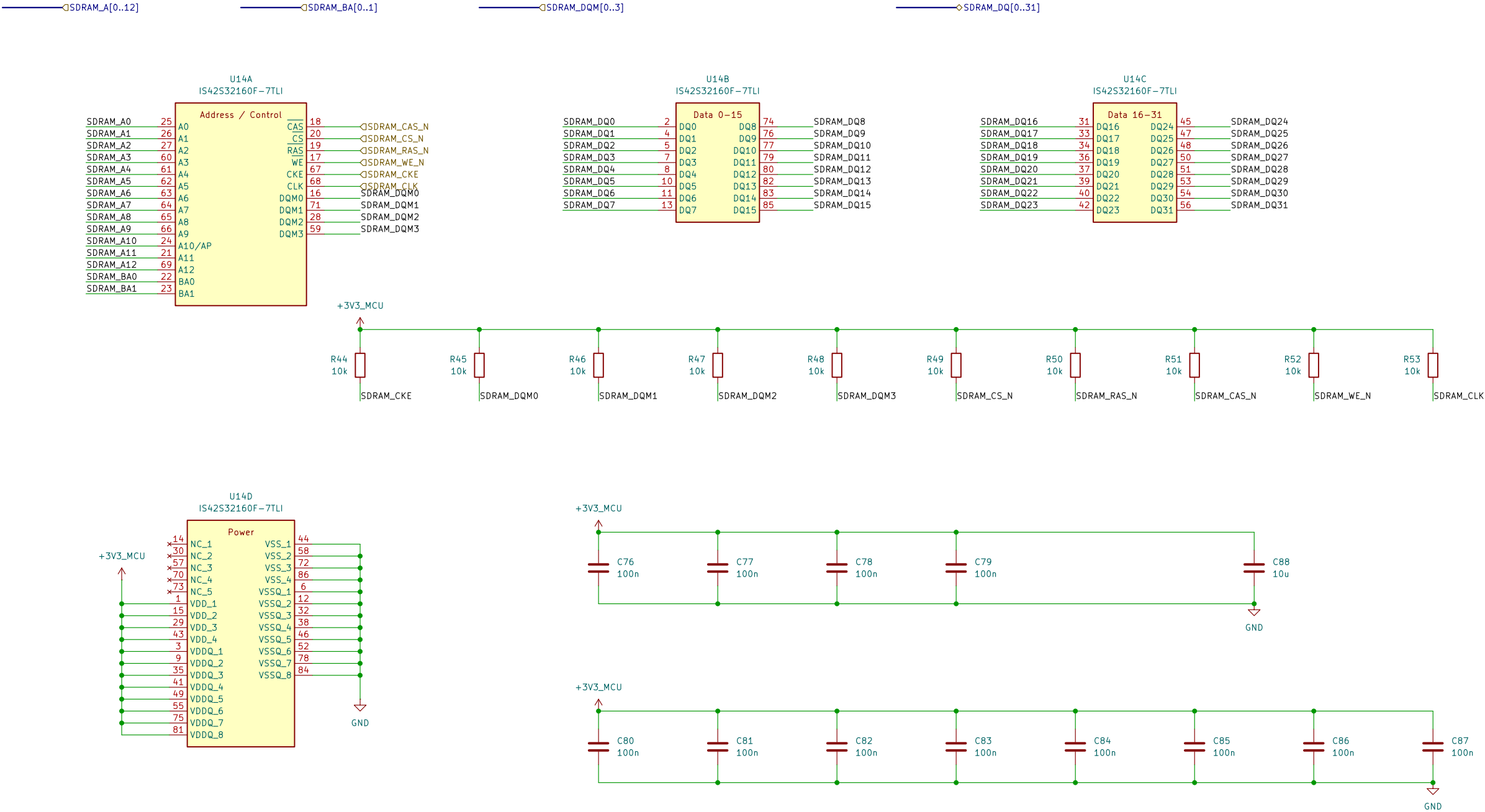
MAIN DIGITAL 3.3 V SUPPLY
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS
U13 TPS63802DLAR; MODE=VIN, fixed PWM.
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.
Vnom=0.500*(1+56k/10k)=3.300V.
Idivider=3.300/(56k+10k)=50uA.
For 100C resistor–temperature change:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vmin=0.495*(1+56k*fLo/(10k*fHi))–100nA*56k*fHi
=3.242044V.
Vmax=0.505*(1+56k*fHi/(10k*fLo))+100nA*56k*fHi
=3.358485V.
Static screen only; ripple, load steps and drift still require validation.
C73: CIN_model=10u*0.90*0.85*0.60=4.590uF >=4uF.
C74+C75: COUT_model=2*22u*0.90*0.85*0.60=20.196uF >=7uF.
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS
Main continuous allocation: 0.750 MCU + 0.500 radio
+ 0.250 SDRAM + 0.100 NOR + 0.050 logic = 1.650A.
These are qualification allocations, not measured maximum currents.
SD, camera, audio, display/touch/frontlight need separate SYS supplies.
Pout=3.358485094*1.65=5.541500W.
VIN>=3.20V allocation; raw trip min=3.263705831V.
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.
At ASSUMED eta=85%: Iin=Pout/(eta*3.20)=2.037316A.
Ploss=Pout*(1/eta–1)=0.977912W.
Reference rise=Ploss*81C/W=79.210859C; NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.400 other = 1.730A.
Radio on would give 2.230A > nominal 2A; keep radio OFF through start/wake.
Largest wake–reference subtotal=1.270+0.400=1.670A.
DCDC figures are reference–only; non–DCDC start and extra charging excluded.
Additional charging: I=Cadditional*dV/dt; qualify the actual ramp.
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.
No externally driven +3V3_MCU: forced PWM can sink reverse current.
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: ../design/power_decoupling.md PWR–003;
source–loss shutdown: ../design/system_power_design.md SYS–007.

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76–C79: 4 x 100 nF for the four VDD pins.
C80–C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44–R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0–7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.